

Coding Practice Set - 03

Loop Logic & Summations :

1. Write a program to calculate the sum of the first N natural numbers.
2. Write a program to calculate the sum of the first N even natural numbers.
3. Write a program to calculate the sum of the first N odd natural numbers.
4. Write a program to calculate the sum of squares of the first N natural numbers.
5. Write a program to calculate the sum of cubes of the first N natural numbers.
6. Write a program to find the sum of all digits of a given number.

Number Processing & Reversals :

7. Write a program to calculate the factorial of a number.
8. Write a program to count digits in a given number.
9. Write a program to reverse a given number.
10. Write a program to check whether a given number is a Palindrome (reads the same backward as forward).
11. Write a program to check whether a given number is an Armstrong number (e.g., $153=1^3+5^3+3^3$).

Mathematical Logic & Number Theory :

- 12. Write a program to check whether a given number is a Prime number or not.**
- 13. Write a program to print all Prime numbers under 100.**
- 14. Write a program to calculate the LCM (Lowest Common Multiple) of two numbers.**
- 15. Write a program to calculate the HCF / GCD (Highest Common Factor / Greatest Common Divisor) of two numbers.**
- 16. Write a program to print the first N terms of the Fibonacci sequence.**